

Applied Mathematics

MP491: Nonlinear Systems (5 ECTS)

This course is an introduction to the analysis of systems of nonlinear Ordinary Differential Equations (ODEs) and Maps.

Taught in Semester(s) II. **Examined** in Semester(s) II.

Workload: 36 hours (24 Lecture hours, 12 Tutorial hours).

Module Learning Outcomes. On successful completion of this module the learner should be able to:

1. Locate and calculate the stability for equilibria in 1-dim ODEs;
 2. Locate and classify bifurcations for equilibria in 1-dim ODEs;
 3. Locate, classify and calculate the stability for equilibria in linear 2-dim systems of ODEs;
 4. Sketch phase-plane portraits about equilibria in linear 2-dim systems of ODEs;
 5. Locate equilibria in nonlinear 2-dim systems of ODEs;
 6. Linearise nonlinear 2-dim systems of ODEs, calculate the linear stability of equilibria and classify equilibria;
 7. Sketch phase-plane portraits of nonlinear 2-dim systems of ODEs using iso-curves;
 8. Analyse 2-dim Hamiltonian systems and sketch their phase-plane portraits;
 9. Locate and classify Hopf bifurcations in nonlinear 2-dim systems of ODEs, and determine the stability of the corresponding limit cycles;
 10. Locate and calculate the stability for fixed points and periodic orbits in 1-dim nonlinear maps;
 11. Locate bifurcations in 1-dim nonlinear maps;
 12. Describe period-doubling cascades to chaos in 1-dim nonlinear maps.
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Indicative Content

This course is concerned with systems of nonlinear Ordinary Differential Equations (ODEs) and Maps. Topics covered include:

1. 1-dimensional differential equations: equilibria, stability, bifurcations;
 2. 2-dimensional linear systems of ODEs: equilibria, stability, phase-plane portraits;
 3. 2-dimensional nonlinear systems of ODEs: equilibria, linearisation, linear stability, phase-plane portraits;
 4. 2-dimensional Hamiltonian systems: equilibria, stability, phase-plane portraits;
 5. Limit cycles: Hopf bifurcations, stability;
 6. 1-dimensional difference equations and maps cycles: fixed points, periodic orbits, stability, bifurcations.
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Module Resources

Strogatz Steven H., Nonlinear Dynamics and Chaos: with Applications to Physics, Biology, Chemistry, and Engineering, Westview Press, 1994,

Kuznetsov Yu.A, Elements of Applied Bifurcation Theory – Third Edition, Springer, 2010.

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